

# LANDSCAPES OF THE CHURCH

## 02 — Food & Working Landscapes

Diocesan and Episcopal Conference layers: overview, ArcGIS Online-ready descriptions, methodology, QC, formulas, workflow, and reproducible code

*LSAP Integral Ecology Atlas | Technical Documentation | 3 October 2026*

## 7. Analysis 02 — Food & Working Landscapes

### 7.1 Diocesan layer overview

|                    |                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Layer name         | LandscapesChurch_02_Dioceses_FoodWorking_V2                                                                                                                                                                                                                                                                                                                      |
| Layer alias        | 02 - Diocesan Food & Working Landscapes                                                                                                                                                                                                                                                                                                                          |
| Scale / features   | Diocese; 2,874 features                                                                                                                                                                                                                                                                                                                                          |
| Description        | Combines cropland and rangeland to describe the extent and internal balance of agricultural and working-land systems within each diocesan analytical geography.                                                                                                                                                                                                  |
| Attribution        | Catholic ecclesiastical geography from the project's versioned diocesan / Episcopal Conference framework; ecclesiastical statistics and leadership enrichment from Holy See statistical publications and Catholic-Hierarchy.org. Environmental land-cover summaries derived from Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagons.      |
| Citation / credits | Burhans et al., Diocesan Boundaries of the Catholic Church [Feature Layer], v1.0 and subsequent v2.0 project revision; Holy See, Annuario Pontificio / Annuario Statisticum Ecclesiae (multiple years); Catholic-Hierarchy.org (accessed Sep 2026); Esri, Global Biodiversity and Conservation Hexagons (Living Atlas; 2026 release / project-accessed version). |
| Analyses           | Cropland + rangeland combined extent; component shares within the combined system; leading working-land type; balance/context; scale-specific ranks retained analytically.                                                                                                                                                                                       |
| Key story fields   | Food/working landscape; Church; scale; summary/comprehensive stories.                                                                                                                                                                                                                                                                                            |
| Status             | Publication-ready                                                                                                                                                                                                                                                                                                                                                |

### 7.2 Episcopal Conference layer overview

|                    |                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Layer name         | LandscapesChurch_02_Conferences_FoodWorking_V2                                                                                                                                                                                                                                                                                                                   |
| Layer alias        | 02 - Episcopal Conference Food & Working Landscapes                                                                                                                                                                                                                                                                                                              |
| Scale / features   | Episcopal Conference; 122 features                                                                                                                                                                                                                                                                                                                               |
| Description        | Summarizes cropland and rangeland across Episcopal Conference geographies to show where food production and working landscapes form a substantial part of the surrounding territorial context.                                                                                                                                                                   |
| Attribution        | Catholic ecclesiastical geography from the project's versioned diocesan / Episcopal Conference framework; ecclesiastical statistics and leadership enrichment from Holy See statistical publications and Catholic-Hierarchy.org. Environmental land-cover summaries derived from Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagons.      |
| Citation / credits | Burhans et al., Diocesan Boundaries of the Catholic Church [Feature Layer], v1.0 and subsequent v2.0 project revision; Holy See, Annuario Pontificio / Annuario Statisticum Ecclesiae (multiple years); Catholic-Hierarchy.org (accessed Sep 2026); Esri, Global Biodiversity and Conservation Hexagons (Living Atlas; 2026 release / project-accessed version). |
| Analyses           | Cropland + rangeland combined extent; internal balance;                                                                                                                                                                                                                                                                                                          |

|                  |                                                                      |
|------------------|----------------------------------------------------------------------|
|                  | Conference, Church, and LSAP contextual stories.                     |
| Key story fields | Food/working landscape; Church; LSAP; summary/comprehensive stories. |
| Status           | Publication-ready                                                    |

## 7.3 Formulae and fields

|                             |                                                                                                                                                                     |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Combined working-land share | $\text{FOODWORK\_PCT} = \text{PCT\_LULC\_Crops} + \text{PCT\_LULC\_Rangeland}$<br>Percentage of the full analysis geography represented by cropland plus rangeland. |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|               |                                                                                                                    |
|---------------|--------------------------------------------------------------------------------------------------------------------|
| Combined area | $\text{FOODWORK\_AREA} = \text{Cropland\_km}^2 + \text{Rangeland\_km}^2$<br>Calculated after unit standardization. |
|---------------|--------------------------------------------------------------------------------------------------------------------|

|                                    |                                                                                                                                                       |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cropland share within working land | $\text{CROP\_SHARE\_FW} = 100 \times \text{Cropland} / (\text{Cropland} + \text{Rangeland})$<br>Internal composition of the food/working-land system. |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                     |                                                                                                                              |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Rangeland share within working land | $\text{RANGE\_SHARE\_FW} = 100 \times \text{Rangeland} / (\text{Cropland} + \text{Rangeland})$<br>Complements CROP_SHARE_FW. |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------|

- FW\_PRIMARY — Cropland or Rangeland, based on the larger component.
- FW\_BALANCE — descriptive measure/context of how balanced the two components are.
- FW\_CLASS — analytical extent/context category.
- CROP\_RANK / CROP\_PCTL and RANGE\_RANK / RANGE\_PCTL — scale-specific comparison fields.

## 7.4 Rank method

Competition rank: rank 1 = highest value. Percentile = percentage of observations at or below the observed value. Diocese and Conference ranking universes are kept separate.

## 7.5 Workflow

1. Standardize source area units.
2. Add cropland and rangeland shares and areas.
3. Calculate each component's share of the combined working-land system.
4. Assign FW\_PRIMARY, FW\_BALANCE, and FW\_CLASS.
5. Calculate separate cropland and rangeland ranks/percentiles by scale.
6. Generate descriptive stories focused on food systems, stewardship, and working landscapes.
7. Run completeness/formula checks and clean the publication schema.

### Retained QC output

FINAL SCHEMA / FEATURE CHECK

Dioceses: 2,874 features

Episcopal Conferences: 122 features

Final diocesan user fields: 58

Final Conference user fields: 200

Ranking universes kept separate by scale.

Competition rank and at-or-below percentile method documented.

# Relevant reproducible code

## A.5 Analysis 02 — Food & Working Landscapes

```
crop = safe(row["PCT_LULC_Crops"])
rangeland = safe(row["PCT_LULC_Rangeland"])

foodwork_pct = crop + rangeland

if foodwork_pct > 0:
    crop_share_fw = 100.0 * crop / foodwork_pct
    range_share_fw = 100.0 * rangeland / foodwork_pct
else:
    crop_share_fw = 0.0
    range_share_fw = 0.0

fw_primary = (
    "Cropland"
    if crop >= rangeland
    else "Rangeland"
)

foodwork_area = (
    safe(row["TOTAL_LULC_Crops"]) +
    safe(row["TOTAL_LULC_Rangeland"])
)
```

## A.10 Safe m<sup>2</sup> → km<sup>2</sup> conversion

```
import statistics

values = []

with arcpy.da.SearchCursor(
    feature_class,
    ["ANALYSIS_AREA"]
) as cursor:

    for (value,) in cursor:
        if value is not None and value > 0:
            values.append(float(value))

median_area = statistics.median(values)

if median_area >= 1_000_000:

    with arcpy.da.UpdateCursor(
        feature_class,
        ["ANALYSIS_AREA"]
    ) as cursor:

        for row in cursor:
            if row[0] is not None:
                row[0] = float(row[0]) / 1_000_000.0
                cursor.updateRow(row)

else:
    print("Area already appears to be in km²; no conversion performed.")
```

## A.11 Rank / percentile reference implementation

```
def competition_rank(values):
    """
    values = iterable of (feature_id, numeric_value)
    Rank 1 = highest.
    Ties receive the same rank; the next rank skips accordingly.
    """
    ordered = sorted(
        values,
        key=lambda x: x[1],
        reverse=True
    )
```

```

)

result = {}
previous = None
current_rank = 0

for position, (feature_id, value) in enumerate(ordered, start=1):
    if previous is None or value != previous:
        current_rank = position

        result[feature_id] = current_rank
        previous = value

    return result

def percentile_at_or_below(all_values, value):
    valid = [v for v in all_values if v is not None]

    if not valid:
        return None

    n = sum(v <= value for v in valid)

    return 100.0 * n / len(valid)

```

## A.16 Publication cleanup pattern

```

# Keep only fields intentionally required for the publication derivative.
KEEP_FIELDS = {
    "BUFFER_ID",
    "Account_Name",
    "Country",
    "Language",
    "Official_website",
    "Sector",
    "BUFFER_AREA",
    "LANDSCAPE_SCALE",
    # ... eight PCT_LULC fields ...
    # ... coverage/mosaic/transition fields ...
    # ... story fields ...
}

delete_fields = [
    f.name
    for f in arcpy.ListFields(feature_class)
    if (
        not f.required and
        f.name not in KEEP_FIELDS
    )
]

if delete_fields:
    arcpy.management.DeleteField(
        feature_class,
        delete_fields
    )

# Always run a final required-field check after deletion.

```

## Sources and attribution

Ecclesiastical geography and Church statistics: LSAP Integral Ecology Atlas / project Catholic jurisdictional framework; Holy See statistical publications; Catholic-Hierarchy.org. Environmental context: Esri ArcGIS Living Atlas Global Biodiversity and Conservation Hexagons and the underlying credited source organizations.

Esri Global Biodiversity and Conservation Hexagons: <https://www.arcgis.com/home/item.html?id=f1bdbcac3c54ad8b5b21f5b17d91e8d>

Esri 2026 update: <https://www.esri.com/arcgis-blog/products/arcgis-living-atlas/sustainable-development/global-biodiversity-and-conservation-hexagons-but-better>

Summarize Within documentation: <https://pro.arcgis.com/en/pro-app/3.6/tool-reference/analysis/summarize-within.htm>